Sai Akhila Buddhi

Python Developer

Location: Florida, USA | **Mail:** Akhila18061999@gmail.com | **Ph.:** 5615715479 | **LinkedIn**

PROFESSIONAL SUMMARY:

Full Stack Python Developer with over 6+ years of experience specializing in architecting, securing and scaling enterprise-grade solutions across high-stakes Financial Services (Digital Payments), E-commerce and Regulated Healthcare (HIPAA/PCI-DSS/SOX) domains.
Expert in designing high-throughput, event-driven Microservices using modern frameworks like FastAPI, Django and DRF & reducing system latency significantly through asynchronous I/O, PostgreSQL tuning and advanced Redis caching.
Proficient in developing interactive, responsive UIs with React/TypeScript, Redux Toolkit and Material-UI. Core expertise spans the entire development lifecycle, including containerization with Docker/Kubernetes (AWS EKS/Helm), IaC (Terraform, CloudFormation), CI/CD automation (Jenkins, GitHub Actions), distributed observability (OpenTelemetry, Prometheus-Grafana) and secure SDLC practices (SAST/SCA, OAuth2, OpenID Connect and JWT).

TECHNICAL SKILLS:

Programming Languages:	Python (3.8–3.11+), TypeScript, JavaScript (ES6+), SQL, HTML5, CSS3
Backend Frameworks & APIs:	FastAPI, Django, Django REST Framework (DRF), Flask, GraphQL, aiohttp, SQLAlchemy, Alembic
Frontend Technologies:	React.js, Redux Toolkit, React Query, Material-UI, Bootstrap, HTML5, CSS3, Responsive UI/UX
Microservices & Architecture:	Event-Driven Architecture, RESTful & GraphQL API Design, Domain-Driven Design (DDD), Containerized Microservices
Databases & Data Modeling:	PostgreSQL, Oracle, MySQL, SQLite, Redis (Caching & Rate Limiting), Alembic (Migrations), Query Optimization, Data Modeling & Indexing
Asynchronous & Messaging Systems:	Apache Kafka, Celery, RabbitMQ, AsyncIO, Multithreading, Background Task Queues
Cloud & Infrastructure:	AWS (EKS, EC2, S3, RDS, CloudFormation, Lambda), DigitalOcean, Terraform, Docker, Kubernetes (Helm), Nginx Reverse Proxy
CI/CD & DevOps Tools:	Jenkins, GitHub Actions, GitLab CI, SonarQube, Black Duck, Terraform, Docker Compose, Helm, Build & Release Automation
Security & Compliance:	OAuth 2.0, OpenID Connect, JWT, IAM Integration, OWASP, PCI-DSS, SOX, HIPAA Compliance, Data Encryption & Sanitization
Monitoring & Observability:	OpenTelemetry, Prometheus, Grafana, ELK Stack, Distributed Tracing, Performance Profiling
Testing & Quality Assurance:	PyTest, Selenium, Postman, Unit & Integration Testing, Test-Driven Development (TDD), API Mocking
Data Engineering & ETL:	Pandas, NumPy, ETL Pipelines, Data Validation & Transformation, Feature Engineering, Batch & Stream Processing
Machine Learning (Applied):	Scikit-learn, Pandas, NumPy (for Recommendation Systems & Predictive Analytics)
Version Control & Collaboration:	Git, GitHub, Bitbucket, GitLab, Code Reviews, Branching Strategies
Build & Dependency Management:	pip, Poetry, Virtualenv, PyPI, npm, Webpack, Babel
System Design & Performance:	API Gateway Optimization, Load Balancing, ORM Optimization, Query Profiling, Asynchronous I/O
Methodologies & Practices:	Agile (Scrum, Kanban), Secure SDLC, DevSecOps, Continuous Integration & Delivery, Code Review & Peer Programming

PROFESSIONAL EXPERIENCE:

JPMorganChase – FL, USA | Python Developer **January 2024 – Present**

Project: Enterprise Digital Payments Modernization Platform

- Built and developed asynchronous REST and GraphQL APIs using FastAPI, Django REST Framework and SQLAlchemy ORM, powering high-throughput digital payments, real-time settlements and internal banking operations.
- Reduced payment transaction latency by 43% through asynchronous I/O processing, PostgreSQL query tuning and optimized ORM layer caching, ensuring sub-second API response times under high transaction volume.
- Built event-driven microservices architecture leveraging Apache Kafka, aiohttp and Docker/Kubernetes (Helm) to enable real-time fraud detection, payment orchestration and reconciliation pipelines.
- Migrated 60% of legacy payment APIs to AWS EKS, implementing containerized deployments with Terraform and AWS CloudFormation, improving deployment efficiency by 38% and enhancing system scalability and fault tolerance.
- Implemented OAuth2, OpenID Connect and JWT-based authentication integrated with enterprise IAM systems, achieving 100% PCI-DSS and SOX compliance across API and data access layers.
- Designed and optimized PostgreSQL and Oracle data models with Alembic-based migrations, improving query execution time by 27% and supporting high-volume payment data analytics.

- Developed interactive React + TypeScript frontends using Redux Toolkit and React Query, delivering responsive dashboards for transaction tracking, audit visibility and real-time risk alerts to internal stakeholders.
- Configured Jenkins and GitHub Actions CI/CD pipelines for automated build, test, lint and multi-environment deployment, enabling zero-downtime releases and consistent code quality across teams.
- Instrumented OpenTelemetry tracing and Prometheus–Grafana monitoring stack, implementing distributed observability and performance optimization across all microservices and data streams.
- Collaborated with DevSecOps and risk compliance teams to integrate automated SAST/SCA scanning (SonarQube, Black Duck) and enforce secure SDLC practices, proactively reducing production vulnerabilities by 30%.

HCLTech – India | Python Developer **October 2020 - November 2022**

Project: ShopSphere – E-Commerce Platform for Multi-Vendor Retail

- Architected RESTful APIs using Django REST Framework to enable seamless multi-vendor product catalog management, shopping cart operations & order fulfillment, processing daily request in high-traffic retail environment
- Designed responsive frontend interfaces with React.js, Redux and Material-UI, delivering device-agnostic user experiences tailored for diverse Indian shoppers, including mobile-first designs for rural and urban users.
- Integrated Razorpay payment gateway for secure, India-specific transactions like UPI and net banking, alongside Stripe for international payments, handling monthly transactions with 99.9% uptime.
- Customized Django Oscar framework to implement modular e-commerce features, including vendor-specific product listings, dynamic pricing and streamlined checkout workflows for multi-vendor operations.
- Optimized PostgreSQL database schemas with advanced indexing techniques, reducing query execution times by 25% to support real-time inventory tracking across multiple vendors.
- Developed a machine learning-based recommendation system using Scikit-learn, boosting user retention by 40% through personalized product suggestions based on Indian consumer behavior patterns.
- Configured Celery for asynchronous task handling, efficiently managing daily background jobs such as order email notifications and automated inventory updates for vendor synchronization.
- Deployed the platform on DigitalOcean infrastructure utilizing Docker containers and Nginx reverse proxy, maintaining 99.95% uptime over two years in a scalable, cost-effective setup for Indian e-commerce demands.
- Established CI/CD pipelines via GitHub Actions, cutting deployment cycles by 30% and facilitating weekly releases of new features like vendor dashboard enhancements.
- Conducted comprehensive unit and integration testing with PyTest and Selenium, attaining 85% code coverage to ensure reliability in critical modules like payment processing and user authentication.
- Secured backend APIs with JWT and OAuth 2.0 protocols, following OWASP security standards to mitigate risks such as CSRF and XSS in a multi-vendor ecosystem prone to high-stakes data exchanges.
- Implemented rate limiting and caching mechanisms with Redis, effectively managing peak loads of concurrent users during festive sales events common in Indian retail.

Accenture – India | Python Developer **August 2018 - September 2020**

Project: Patient Data Management System (PDMS)

- Developed a comprehensive Patient Data Management System (PDMS) to streamline patient information storage, retrieval and analysis, ensuring adherence to HIPAA compliance standards.
- Designed and developed secure RESTful APIs using Django REST Framework (DRF) for access to patient data across multiple modules.
- Implemented user authentication and authorization using JWT (JSON Web Tokens) to maintain data security and regulatory compliance.
- Developed frontend application components in React.js, creating an intuitive user interface for patient data entry, search and reporting.
- Utilized PostgreSQL as the primary RDBMS for patient data storage, with schema optimization for data integrity, scalability and cost reduction.
- Improved data retrieval speed by 35% through the effective implementation of Redis caching for frequently accessed patient records.
- Integrated HL7 interfaces for seamless interoperability with external Electronic Health Record (EHR) systems, supporting real-time data exchange.
- Implemented robust data validation and sanitization mechanisms to prevent security breaches and ensure high-quality patient information.
- Developed ETL pipelines using Pandas to process and analyze large-scale patient datasets for clinical and operational insights.
- Deployed and maintained the PDMS application on AWS (EC2, S3 and RDS), leveraging Docker containers and Nginx for scalable, high-availability solutions.
- Established continuous integration and deployment (CI/CD) pipelines using Jenkins, automating build, test and deployment workflows for rapid delivery.
- Conducted comprehensive testing and debugging using PyTest and Postman, ensuring reliability, functionality and compliance of the PDMS application.

EDUCATION:

Master of Computer and Information Science - Florida Atlantic University, Florida, USA
Bachelor of Computer Science - Malla Reddy Engineering College for Women, India